

Orumbek Alinur

SYSTEMS - NUCLEAR ENGINEER - ROBOTICS LEAD



I don't just build robots; I engineer solutions for extreme environments — from competitive FTC/FGC arenas to conceptual Venus atmospheric habitats and research at the INP.

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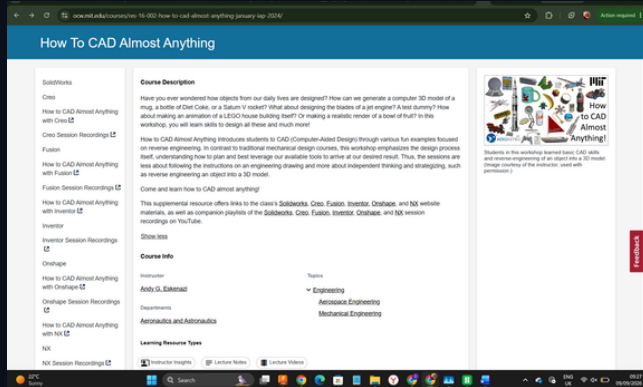
+7 776 524 2371

Abstraction about me

[Your Photo]

Background & Evolution

- **Community:** Active volunteer with the League of Volunteers of Kazakhstan.



- **Self-Taught Execution:** Mastered full-assembly CAD (Fusion 360) via MIT OpenCourseWare under severe constraints.
- **The Pivot:** Shifted focus from humanities & political goals at Abai National School to hardcore systems engineering.



Core Toolkit

Fusion 360

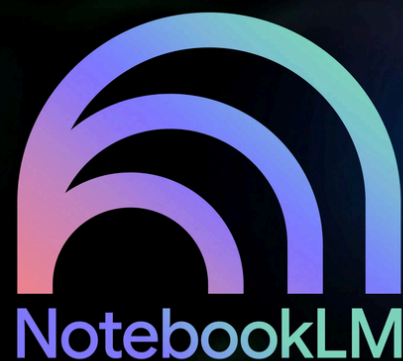
Rapid Prototyping

Aerothermodynamics

ISRU Architecture

Python

Crisis Management



ARENA & HARDWARE // COMPETITIONS

Competitive Edge: Lead CAD Engineer and mechanical builder across FTC & FGC seasons. Survived high- intensity crisis management, build-season bottlenecks, and overnight hardware iterations.

VISUAL DOCUMENTATION

Awards & Badges with my team Antares 24935



CERTIFICATE

Think Award 2nd Place

PRESENTED TO

Orumbek Alinur Kuanyshuly

MENTOR

Mukhamedzhanova Akmaral, Mustafayeva Nazerke

Murzakhmetov A.K.
Director of the USTEM Foundation

Butakhanov Z.
Director of Robotics and Engineering Association of the Republic of Uzbekistan

Scan to verify

EU funded "Last-mile Connectivity to Promote Digital-Green Education Opportunities (Connectivity)" Project

CERTIFICATE

Reach Award Winner

PRESENTED TO

Orumbek Alinur Quanyshuly

MENTOR

Alibekkyzy Aijan, Myxamedzhanova Akmaral

Murzakhmetov A.K.
Director of USTEM Foundation

Scan to verify

CERTIFICATE

Sustain Award 3rd Place

PRESENTED TO

Orumbek Alinur Quanyshuly

MENTOR

Mukhamedzhanova Akmaral, Alibekkyzy Aizhan

Murzakhmetov A.K.
Director of USTEM Foundation

Scan to verify

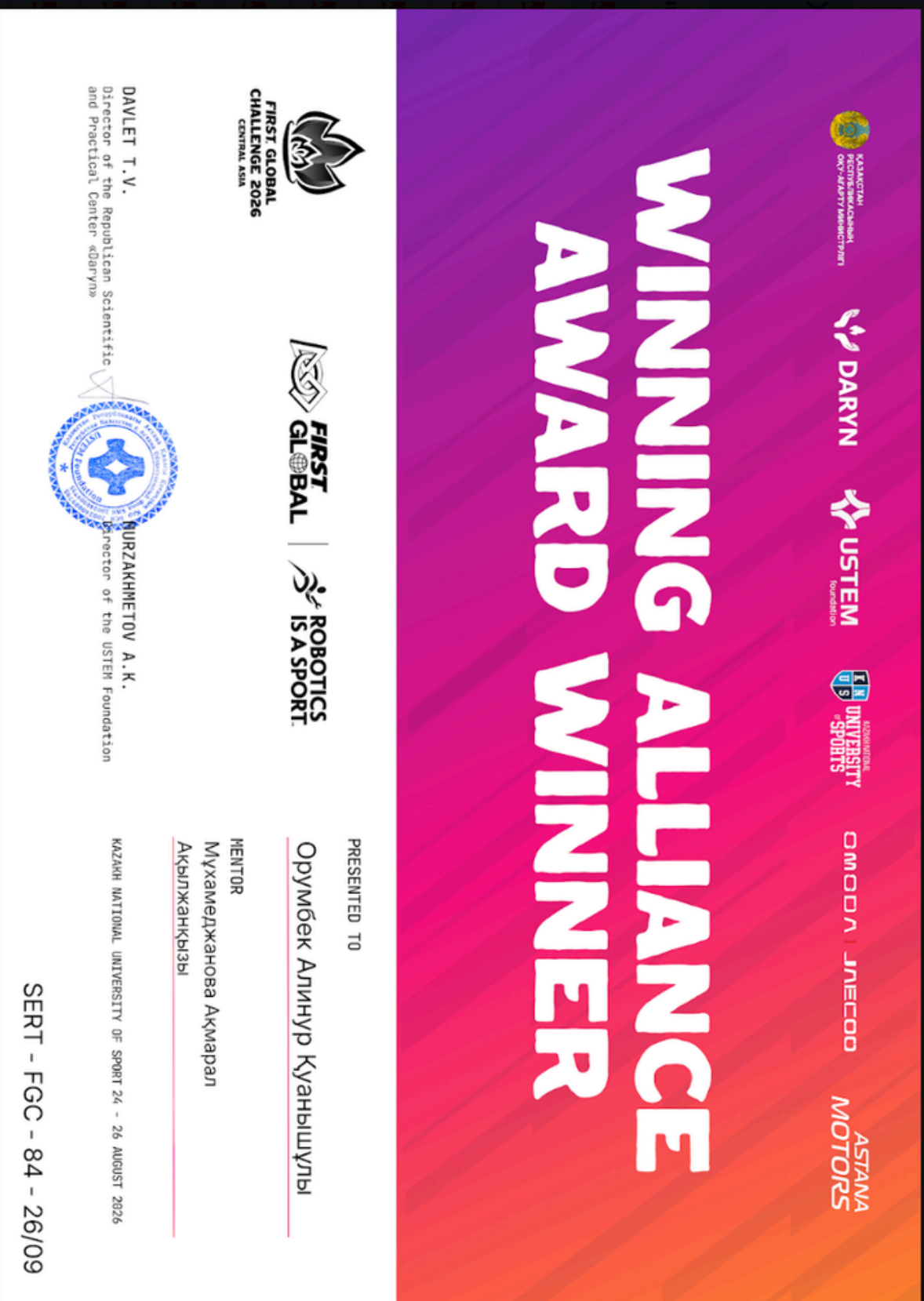
FGC with my team Abai bol



FGC Central Asia — Mechanical Build Lead

- Built and assembled competition robot from scratch without CAD, working directly with hardware under tight deadlines.
- Made real-time design decisions on mechanism placement, structure, and wiring to ensure functionality during competition.
- Collaborated with team to troubleshoot and repair robot between matches.
- Result: Successfully competed at FGC Central Asia Won 1st place as part of alliance at FGC CENTRAL ASIA, 2026" Skills Challenge 3rd Place → "Placed 3rd in individual robot driving & programming skills,

FGC with my team Abai bol





INTERNATIONAL UNITY AWARD 2ND PLACE



FIRST GLOBAL
CHALLENGE 2026
CENTRAL ASIA



FIRST
GLOBAL



ROBOTICS
IS A SPORT

PRESENTED TO

Орумбек Алдинур Куанышұлы

MENTOR

Мұхамеджанова Ақмарал Ақылжанқызы

DAVLET T. V.

Director of the Republican Scientific
and Practical Center «Daryn»



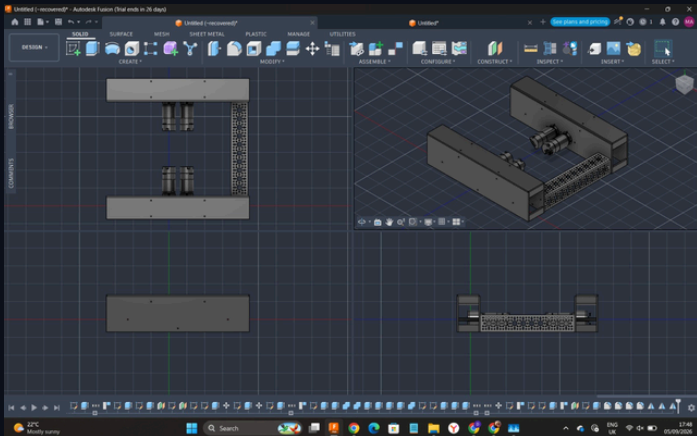
MURZAKHMETOV A. K.

Director of the USTEM Foundation

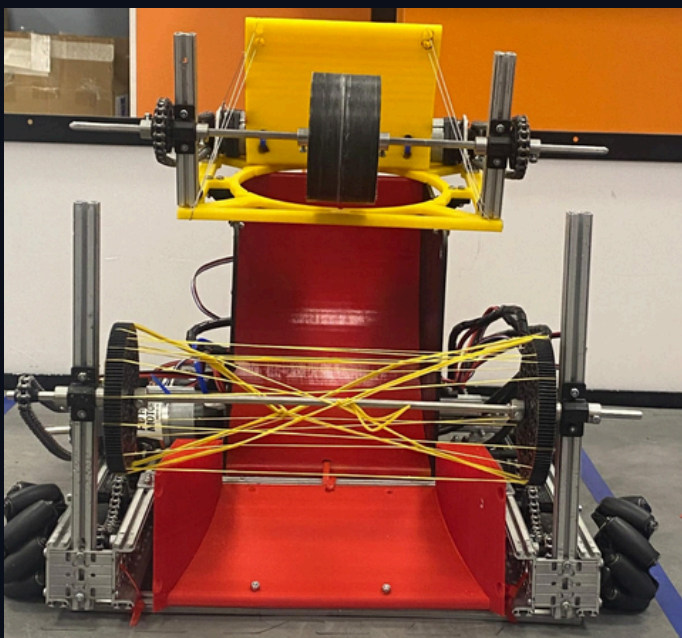
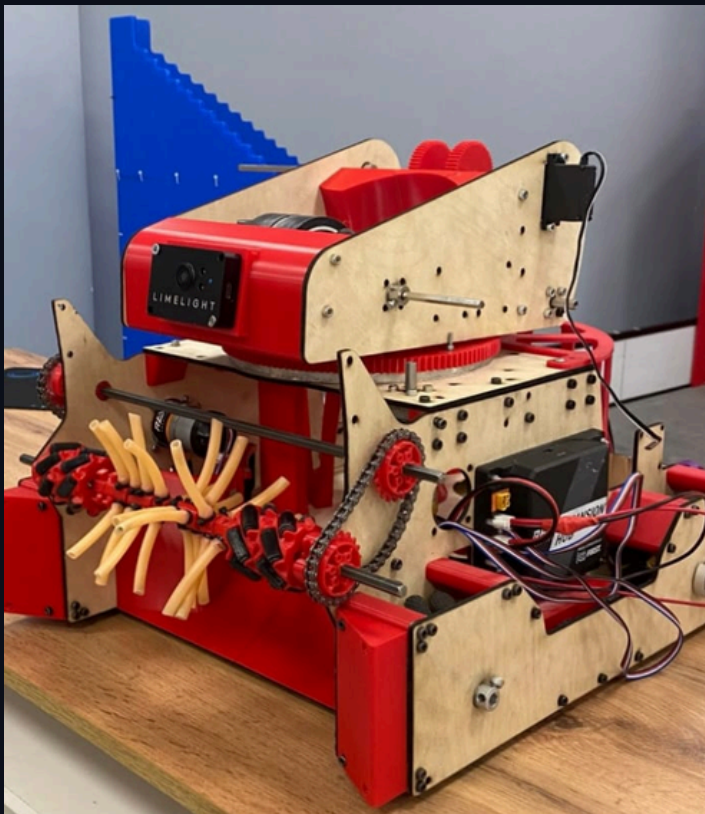
KAZAKH NATIONAL UNIVERSITY OF SPORT 24 - 26 AUGUST 2026

SERT - FGC - 79 - 26/09

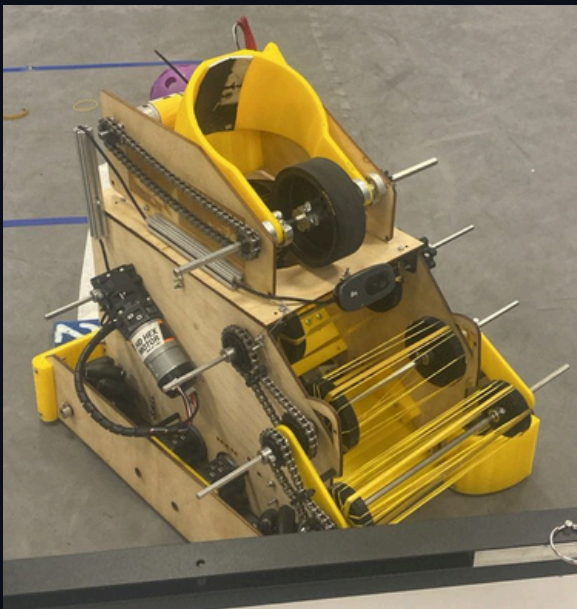
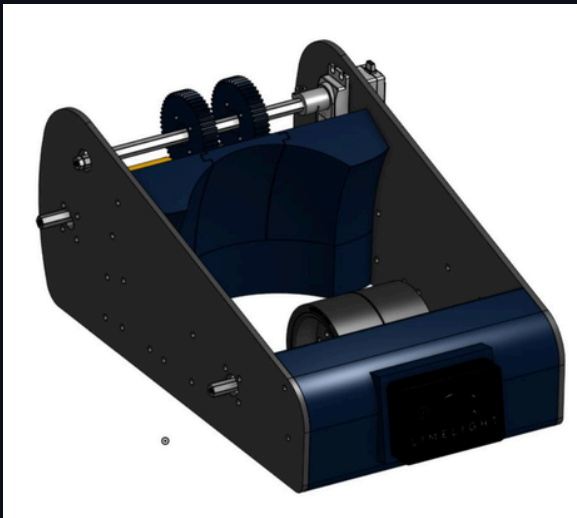
Fusion 360 CAD Renders robot reveal



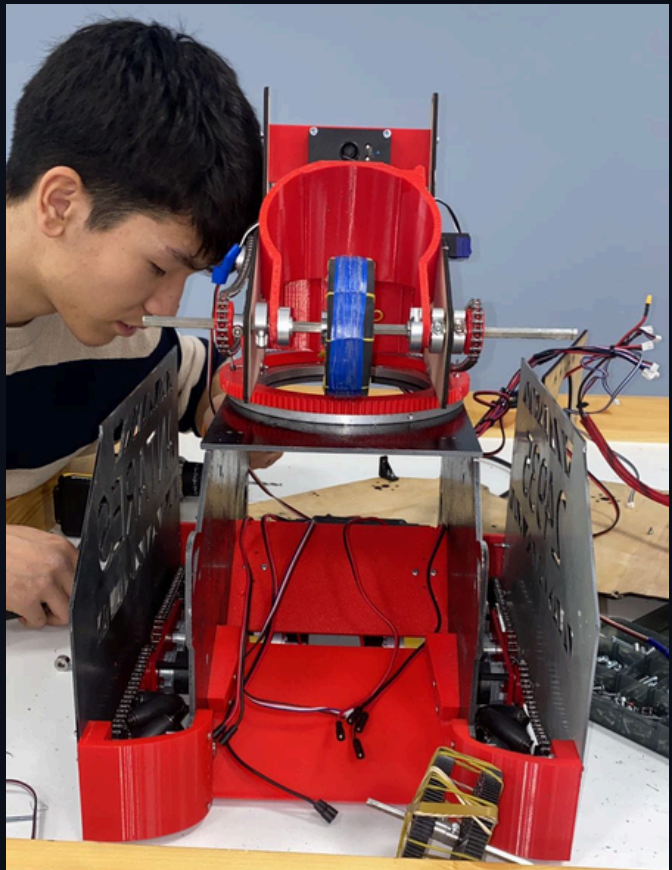
Robot Build & Workshop



Fusion 360 CAD Renders robot reveal

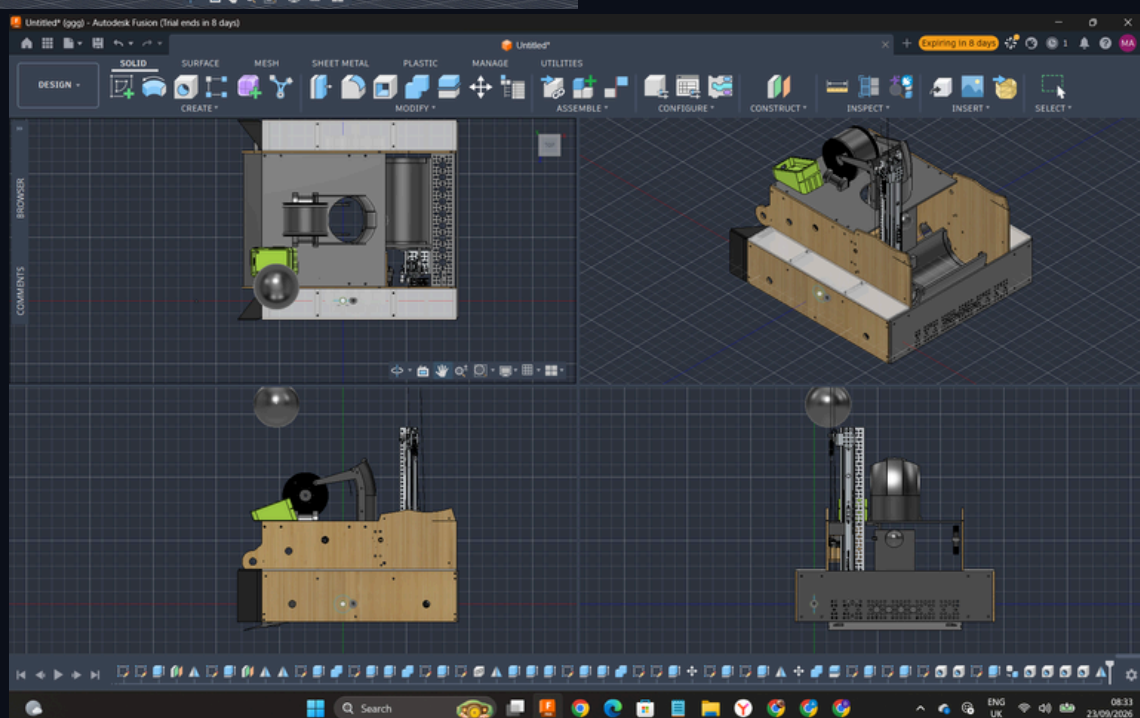
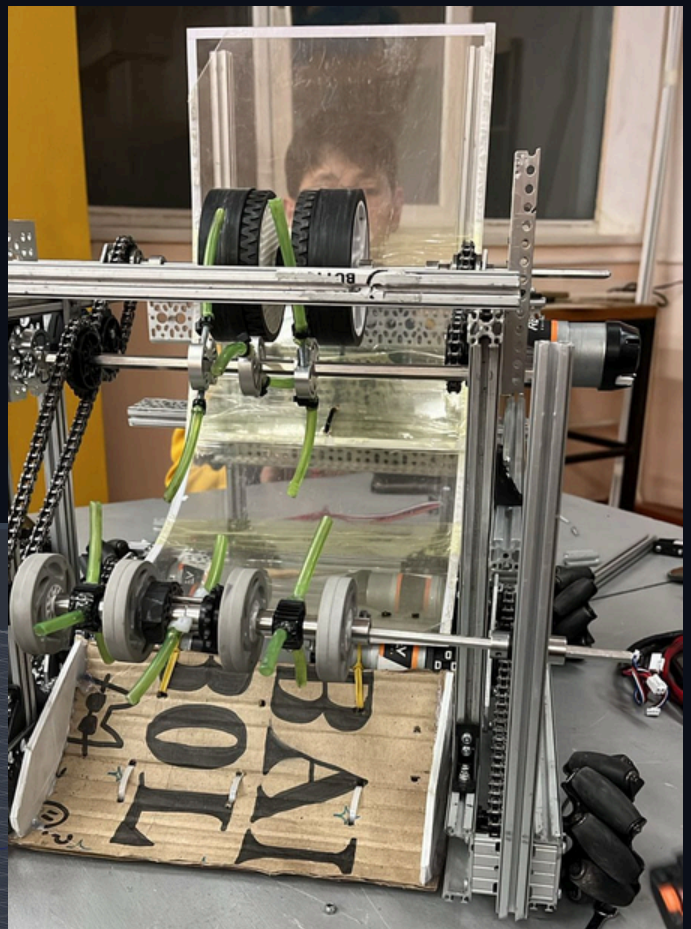
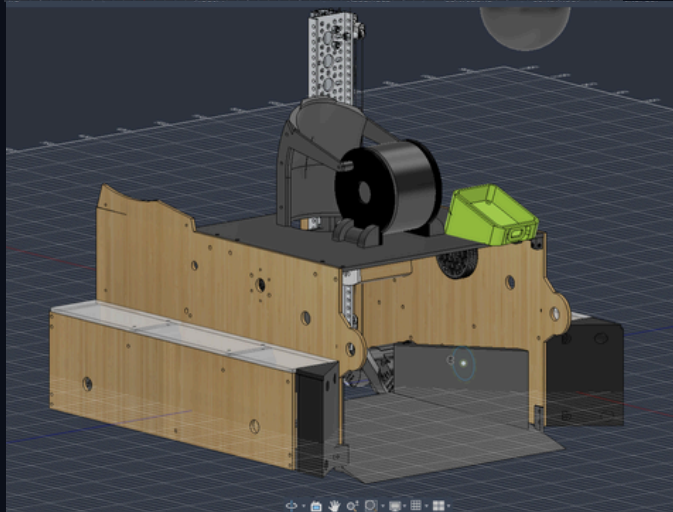
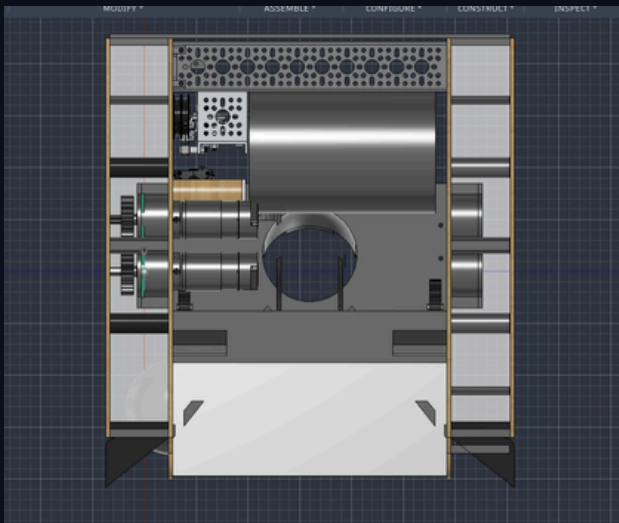


Robot Build & Workshop



Fusion 360 CAD Renders robot reveal

Robot Build & Workshop



FEATURED PROJECT // AEGIS-VENUS (AEGIS-V)



Atmospheric Habitat Framework: A multi-phase systems engineering roadmap inspired by NASA's HAVOC concept. Targets the upper atmosphere (~50 km altitude) of Venus to bypass surface hell (~460°C, 92 atm), utilizing standard air as a natural lifting gas.

Subsystems:

- **EDI Sequence:** Entry, Descent & Inflation for hypersonic deceleration.
- **TPS:** Advanced flexible thermal protection systems.

Autonomy & ISRU:

- Automated propellant generation.
- On-site 3D printing & long-term life-support autonomy.

